Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

Supplement to: Horn L, Mansfield AS, Szczęsna A, et al. First-line atezolizumab plus chemotherapy in extensive-stage small-cell lung cancer. N Engl J Med 2018;379:2220-9. DOI: 10.1056/NEJMoa1809064

SUPPLEMENTARY APPENDIX

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Supplement to: Horn L, Mansfield AS, Szczęsna A, et al. Atezolizumab Plus Chemotherapy in First-Line Extensive-Stage SCLC

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The following investigators participated in the IMpower133 study:

Australia – R. De Boer*, M. Barnet, M. Boyer, D. Herath, B. Hughes*, P.-L. Inglis, S. Kao*, Z. Lwin, K. Morris, B. Solomon, L. Tan, A. Thai, A. Zhang

Austria – E. Brehm, O. Burghuber*, N. Dworan, S. Gasser, S. Hartl*, K. Hergan, M. Hochmair, F. Huemer, I. Kapfhammer, R. Koger, A. Koköfer, H. Kratochwill, R. Kropfmüller, B. Lamprecht*, R. Mikes, A. Mohn-Staudner, C. Pirker, P. Porsch, B. Rohrer, I. Schmatzer, E. Schober, E. Stiefsohn, M. Studnicka*

Brazil – L. Andrade*, L. H. Araujo*, S. Azevedo, A. C. Borghetti, L. Brust*, C. Campos*, G. Castro Junior*, J. Coelho, M. Costa, A. Dettino*, A. Faccio*, M. Felizola, F. A. Franke*, F. Fujiki, C. Gil*, P. E. Liedke, V. Lopes*, M. Mak, R. Martins, C. Mesquita, L. Moura, R. Pereira*, F. Peria*, F. Roitberg, F. Santini, G. Santos Borges*, L. A. Schlittler*, N. Schuh, R. Seewald, L. Sousa, T. Takagaki, M. Tenorio Silva, G. Werutsky*

Chile – O. Aren*, R. Conejeros, F. Orlandi*, I. Quijada, M. P. Saavedra, E. Yanez*, P. Yañez

China – Y. Chen, Y. Cheng*, H. Cui, H. Ding, N. Li, Z. Li, J. Liu, Y. Liu, H. Xia, Y. Xin, N. Xu, Y. Yang, J. Zhu

Czech Republic – M. Bartouskova, V. Benysek, M. Cernovska, L. Havel*, M. Hudcova, R. Kovar, B. Melichar*, P. Opalka*, N. Pauk, H. Studentova, D. Vitaskova

France – D. Debieuvre*, M. Baciuchka Palmaro, F. Barlesi, P. E. Brachet, M. Cabart, F. Chomy, E. Coquan, A. Cortot, S. Cousin*, X. Dhalluin, P. Do, C. Dubos Arvis, C. Fournier de Lesquen, M. E. Garcia, C. Gauvain, R. Gervais*, L. Greillier*, B. Ladam, C. Mascaux, E. Meriaux, C. Munck, A. Scherpereel*, C. Segura, P. Tomasini, M. Toulmonde, M. C. Willemin

Germany – M. Ambold, T. Behlendorf, H. Bischoff*, F. Bozorgmehr, T. Duell, H. Grosch, D. Heigener, A. Hermes, Y. Hofmann, J. Karin, M. Knopf, C. Küster, M. Möller, M. Reck*, N. Reinmuth*, S. Reitmeier, A. Rittmeyer*, W. Schütte*, B. Storbeck, D. Wagner-Hug, B. Zwerger

Greece – R. Argiriadi, P. Boura, A. Charpidou, T. Frantzi, M. Gerolympou, T. Gkaravela, S. Gkoura, H. Kalofonos*, E. Res, E. Samantas*, J. Sgouros, G. Stamoulis, K. Syrigos*, S. Tsimpoukis, L. Vathiotis, A. Visvikis

Hong Kong – C. K. Cheng*, K. C. Lam, Y. T. M. Lam, WY. Lee, T. S. K. Mok*, A. Poon, T. Shum

Hungary – T. Borbely, Z. Kozma, Z. Mark*, S. Aleid, E. Hidvegi, Z. Kelemen, G. Ostoros*, M. Pap, T. Kardos, M. Szilasi*, L. Kunos, G. Losonczy*, L. Tamasi

Italy – S. Buti, G. Carteni*, C. Catania, A. Chella*, F. De Marinis*, E. Del Signore, C. Di Micco, F. Facchinetti, I. Fioroni, M. Garassino*, F. Gelsomino, M. Imbimbo, G. Lo Russo, M. Lucchesi, E. Maiello*, A. Passaro, I. Petrini, C. Pochesci, C. Proto, G. Puppo, G. Rossi, D. Signorelli, G. Spitaleri, M. Tiseo*, G. Tonini*, M. Vitali, N. Zilembo

Japan – Y. Amino, Y. Abe, Y. Adachi, M. Aga, T. Aiba, H. Akamatsu*, K. Akamatsu, M. Aso, S. Atagi*, K. Azuma, Y. Domeki, T. Enomoto, K. Furuta, K. Furuta, K. Furuuchi, K. Hanaoka, K. Haratani, S. Harita, T. Hasegawa, H. Hayashi*, A. Hayata, K. Hirano, R. Hiraoka, N. Hirata, E. Hirokawa, Y. Honda, A. Horiike, Y. Hosomi, T. Imabayashi, Y. Inagaki, T. Ishida, A. Ito, E. Iwama, T. Iwasa, R. Kagami, T. Kagawa, K. Kanai, Y. Kaneko, K. Katayama, R. Kato, T. Kato, Y. Kato, T. Kato*, R. Katsuda, H. Kawakami, T. Kawamura, T. Kawamura, S. Kawana, Y. Kawashima, H. Kenmotsu, Y. Kimura, S. Kitazono, H. Kobayashi, T. Kobayashi, Y. Koh, S. Kohno, R. Kominami, T. Kondo, M. Kotake, T. Kozuka, Y. Kurahara, F. Kurimoto, N. Mamesaya, Y. Matsuda, S. Minomo, K. Miyake, E. Miyawaki, Y. Mizumori, S. Mizuno, H. Mizutani, M. Morise*, H. Murakami, E. Murakami, Y. Murakami, T. Naito, K. Nakagawa, Y. Nakahara*, Y. Nakahara, A. Nakamura, M. Nakanishi, Y. Nakanishi, T. Nakano, K. Nakao, K. Nakashima, Y. Naoki, T. Nishihara, M. Nishio*, N. Nishioka, Y. Nonagase, I. Okamoto*, T. Okamura, K. Okishio, Y. Okuda, Y. Okuma*, T. Okuno, N. Okura, S. Omori, N. Onda, Y. Onishi, A. Ono, H. Ono, K. Otsubo, K. Otsuka, S. Saijo, S. Saito, R. Saito, H. Sakai*, H. Sakai, S. Sasaki, H. Shimizu, T. Shimokawaji, S. Sonobe, S. Sugawara*, T. Sugimoto, J. Sugisaka, S. Suzuki, K. Suzuki, M. Suzuki, T. Takahama, T. Takahashi*, K. Takayama*, M. Takeda, N. Takegawa, R. Takei, N. Takeuchi, T. Takimoto, A. Tamiya, N. Tamiya, T. Tanaka, T. Tanaka, K. Tanaka, K. Tanaka, Y. Taniguchi, K. Tanimura, S. Teraoka, K. Terayama, Y. Toi, S. Tokura, H. Tsukamoto, K. Tsurumi, J. Tsurutani, K. Uchibori, J. Uchino, H. Ueda, S. Uematsu, R. Usui, K. Wakuda, S. Watanabe, K. Watanabe, K. Yamada, T. Yamada, S. Yamanda, Y. Yamane, A. Yamazaki, N. Yanagitani, T. Yokose, T. Yokoyama*, M. Yomota, K. Yonesaka, Y. Yoneshima, T. Yoshida, T. Yoshizawa

Republic of Korea – J.-S. Ahn, M.-J. Ahn*, M. Choi, D. Heo, B. Keam, D.-W. Kim*, D.-W. Kim, M. Kim, S.-W. Kim*, S. H. Kim, Y.-J. Kim, D.W. Lee, D. H. Lee, J.-S. Lee*, K. Park, H.K. Park, J. M. Sun, S.K. Yoon

Mexico – J. A. Alatorre Alexander*, D. Añorve, C. Deneken, M. I. Enriquez Aceves*, L. V. Garcia, A. Güemes, E. Macedo*, F. Martinez, A. Ruiz Garcia

Norway – Ø. Fløtten*, B. Grønberg*, A. Helland*

Poland – S. Andrzejewska, P. Badurak, M. Bryl*, R. Dziadziuszko*, J. Dzikowska-Dalek, J. Gajda-Wrzosek, I. Golda-Gocka, M. Gorska-Szymot, O. Grabowska, A. Guzowska, E. Huczynska- Szubert, A. Janowicz-Zebrowska, P. Jaskiewicz, J. Jassem, E. Kalinka-Warzocha, A. Kazarnowicz*, M. Knetki-Wróblewska, J. Kolb-Sielecki, K. Konopa, A. Kowalczyk, D. Kowalski, M. Krzakowski*, M. Kukulska, M. Mazur-Roszak, A. Niedziolka, B. Nikisch, E. Nowak, M. Nowak-Glanc, K. Oniszh, K. Pieniuk-Faszynska, A. Piorek, A. Pluzanski, T. Sawicki, M. Smorczewska, K. Sosinska-Mielcarek, K. Stankiewicz, W. Sucharski, I. Symonowicz*, P. Szambora, A. Szczesna*, M. Torous, K. Winiarczyk, A. Wrona, K. Zajda, M. Zukowska

Russian Federation – S. Bagrova, A. Danilova, A. Filippov*, A. Gladkova, V. Gorbunova*, N. Karaseva*, A. Klischuk, R. Kuliev, A. Kuzminov, E. Levchenko*, N. Levchenko, O. Mamontov, G. Mironova, S. Nikolaeva, V. Nyashin, A. Osheychik, I. Profatilo, Y. Shamanskaya, A. Smolin*, D. Stroyakovskii*, A. Yurchenkov, R. Yurin, A. Zolotarev

Serbia – Z. Andric*, D. Bokan, D. Bursac, M. Cekic, J. Djedovic, I. Djordjevic, S. Djordjevic, M. Grujic, B. Ilic*, D. Jovanovic, S. Kostic, T. Kovacevic, V. Mihailović Vučinić, N. Milutinovic, Z. Murtezani, B. Perin*, S. Popevic*, M. Rancic*, N. Samardzic, T. Sarcev, G. Stojanovic, V. Stojsic, Z. Uskokovic Stefanovic, B. Zaric

Spain – M. Alonso, T. Alonso Gordoa, M. A. Berciano, R. Bernabe Caro, M. Cazorla López, S. M. Cedres Perez, M. Cobo Dols, P. Cruz, J. De Castro Carpeno*, A. Fernández Ruiz, M. J. Flor Oncala, P. Gajate Borau, M. N. Galán, I. Gallego, P. Garrido Lopez, A. Gómez Rueda, P. Gomez Salcedo, V. Gutierrez, D. Isla Casado*, B. Jimenez, A. Martinez, V. Martinez Marin, A. Navarro*, M. E. Olmedo*, C. Quero Blanco, I. Ramos Garcia, J . Remon Masip, M. I. Saez, A. Sanchez, A. Sanchez, A. Soria Rivas, J. M. Trigo Perez

Taiwan – W.-C. Chang, H.-S. Chao, K.-Y. Chen, C.-L. Chiang, C.- H. Chiu*, Y.-F. Fang, C.-C. Ho, C.-L. Hsu, S.-L. Lai, J.-H. Lee, S.-H. Li, W.-Y. Liao, C.-C. Lin, Y.-L. Lin, Z.-Z. Lin, C.-Y. Liu, Y.-H. Luo, T.-H. Shiao, R.-F. Shie, J.-Y. Shih, T.-H. Tsai, C.-M. Tsai, C.-L. Wang, C.-Y. Yang, C.-H. J. Yang, C.-T. Yang*, C.-J. Yu*

Turkey – M. Erman*

United Kingdom – S. D. Aung, G. Brand, R. Califano*, J. Conibear*, V. Ford, F. Gomes, E. Karapanagiotou*, R. Lal, L. Lim, A. Montes, M. Mooney, M. Nicolson*, C. Powell, J. Spicer, C. Stillwell, B. Szabados, E. Toy*, E. Valentine Crowe, V. Vigneswaran*

United States – A. Adjei, E. Ahn, C. Alemany, J. Baither, M. Bajaj, M. Batus*, D. Benton, A. Bickar, S. Blau, A. Blom, P. Bonomi, D. Bowen, F. Braiteh*, M. Bregenzer, A. Brown, J. Bruce, K. Bumgarner, B. Callaway, T. Campbell, E. Casper, J. M. Chaves, M. Cogan, M. Cole-Almgren, E. Crane*, E. T. Creagan, A. Damon, D. Daniel*, K. Davalli, M. de Oliveira, D. DeShazo, M. Dugan, K. Ebrahim, D. Evans, J. Fairfield, B. Fernbach, M. Fidler, P. Fishkin, B. Gershenhorn*, G. J. Gerstner, G. Giuseppe, J. Goldschmidt-Jr*, J. Hamm*, E. Hanisch, R. Harris*, D. Haycraft, C. Hocum, L. Horn*, M. Hussein*, C. Ilyas, R. Induru*, K. Jacks, M. Johns, M. Johnson*, S. Joseph, G. Kashef, T. J. Keating, G. Kennedy, S. Khan, E. Kingsley, E. Kirshner*, J. Kloss, D. Knoll, R. Korst, S. Kurian, L. LaBrecque, H. Larder, T. Leal*, J. Lee, N. Le-Lindqwister, K. Leventakos, S. Liu*, G. MacVicar*, A. Magnoli, A. Mansfield*, R. S. Marks, L. Martin, R. McCroskey, S. L. McCune*, K. McGill, T. Mekhail*, K. Merrill, S. Miles, K. Miller, J. R. Molina, S. Mowry, A. Mullery, R. Neelam, R. Neider, G. Obara, R. Parikh, I. J. Percent*, E. Pisick, H. K. Prasad, X. Qin, T. Rakowski, S. Rao*, J. Readling, N. Rehman, K. Saifuddin, S. Sambidi, W. Samlowski, S. Samu-Arce, H. Sanatinia, J. D. Sanchez, D. Schaffer, F. Senecal, T. W. Sirois, M. Socinski, S. Southwick, K. Spalik, A. I.

Spira*, D. Subramaniam, J. Suh, R. Suppiah, E. Teplinsky, C. Thomas*, J. Thompson, A. Traynor, S. Trupti, P. Varadarajan, A. Veatch*, B. Vicuna, L. Villaruz*, B. Vincent, N. Vogelzang, P. Waterman, T. F. Weisberg, N. West, K. Wood, M .Wrona, M. Yalamanchili

*Indicates Principal Investigators.

Supplementary Methods

Inclusion/exclusion criteria

Inclusion Criteria

Patients must meet all of the following criteria to be eligible for study entry:

- Signed Informed Consent Form
- Male or female, 18 years of age or older
- ECOG performance status of 0 or 1
- Histologically or cytologically confirmed ES-SCLC (per the Veterans Administration Lung Study Group (VALG) staging system
- No prior systemic treatment for ES-SCLC
- Patients who have received prior chemoradiotherapy for limited-stage SCLC must have been treated with curative intent and experienced a treatment-free interval of at least 6 months since last chemotherapy, radiotherapy, or chemoradiotherapy cycle from diagnosis of extensive-stage SCLC
- Patients with a history of treated asymptomatic CNS metastases are eligible, provided they meet all of the following criteria:
    - Only supratentorial and cerebellar metastases allowed (i.e., no metastases to midbrain, pons, medulla or spinal cord)
    - No ongoing requirement for corticosteroids as therapy for CNS disease
    - No evidence of interim progression between the completion of CNS-directed therapy and randomization
    - Patients with new asymptomatic CNS metastases detected at the screening scan must receive radiation therapy and/or surgery for CNS metastases. Following treatment, these patients may then be eligible without the need for an additional brain scan prior to randomization, if all other criteria are met.
- Measurable disease, as defined by RECIST v1.1
    - Previously irradiated lesions can only be considered as measurable disease if disease progression has been unequivocally documented at that site since radiation and the previously irradiated lesion is not the only site of disease.
- Adequate hematologic and end organ function, defined by the following laboratory results obtained within 14 days prior to randomization:
    - ANC ≥1500 cells/µL without granulocyte colony-stimulating factor support
    - Lymphocyte count ≥500/µL
    - Platelet count ≥100,000/µL without transfusion
    - Hemoglobin ≥9.0 g/dL

        - Patients may be transfused to meet this criterion.
    - INR or aPTT ≤1.5 × upper limit of normal (ULN)
        - This applies only to patients who are not receiving therapeutic anticoagulation; patients receiving therapeutic anticoagulation should be on a stable dose.
    - AST, ALT, and alkaline phosphatase ≤2.5 × ULN, with the following exceptions:
        - Patients with documented liver metastases: AST and/or ALT ≤5 × ULN
        - Patients with documented liver or bone metastases: alkaline phosphatase ≤5 × ULN
    - Serum bilirubin ≤1.25 × ULN
        - Patients with known Gilbert disease who have serum bilirubin level ≤3 × ULN may be enrolled.
    - Serum creatinine ≤1.5 × ULN
- Patients must submit a pre-treatment tumor tissue sample. Any available tumor tissue sample can be submitted. The tissue sample should be submitted before or within 4 weeks after randomization; however, patients may be enrolled into the study before the pre-treatment tumor tissue sample is submitted.
- For women of childbearing potential: agreement to remain abstinent (refrain from heterosexual intercourse) or use contraceptive methods that result in a failure rate of <1% per year during the treatment period and for at least 5 months after the last dose of study treatment.
    - A woman is considered to be of childbearing potential if she is postmenarcheal, has not reached a postmenopausal state (≥12 continuous months of amenorrhea with no identified cause other than menopause), and has not undergone surgical sterilization (removal of ovaries and/or uterus).
    - Examples of contraceptive methods with a failure rate of <1% per year include bilateral tubal ligation, male sterilization, established, proper use of hormonal contraceptives that inhibit ovulation, hormone-releasing intrauterine devices, and copper intrauterine devices.
    - The reliability of sexual abstinence should be evaluated in relation to the duration of the clinical trial and the preferred and usual lifestyle of the patient. Periodic abstinence (e.g., calendar, ovulation, symptothermal, or postovulation methods) and withdrawal are not acceptable methods of contraception.

- For men: agreement to remain abstinent (refrain from heterosexual intercourse) or use contraceptive measures, as defined below:
    - With female partners of childbearing potential or pregnant female partners, men must remain abstinent or use a condom during treatment with chemotherapy (i.e., carboplatin and etoposide) and for at least 6 months after the last dose of chemotherapy to avoid exposing the embryo.
    - The reliability of sexual abstinence should be evaluated in relation to the duration of the clinical study and the preferred and usual lifestyle of the patient. Periodic abstinence (e.g., calendar, ovulation, symptothermal, or postovulation methods) and withdrawal are not acceptable methods of contraception.

<u>Exclusion Criteria</u>

Patients who meet any of the following criteria will be excluded from study entry:

- Active or untreated CNS metastases as determined by computed tomography (CT) or magnetic resonance imaging (MRI) evaluation during screening and prior radiographic assessments
- Spinal cord compression not definitively treated with surgery and/or radiation or previously diagnosed and treated spinal cord compression without evidence that disease has been clinically stable for ≥1 week prior to randomization
- Leptomeningeal disease
- Uncontrolled pleural effusion, pericardial effusion, or ascites requiring recurrent drainage procedures (once monthly or more frequently)
    - Patients with indwelling catheters (e.g., PleurX®) are allowed regardless of drainage frequency.
- Uncontrolled or symptomatic hypercalcemia
    - Patients who are receiving denosumab prior to randomization must be willing and eligible to discontinue its use and replace it with a bisphosphonate while in the study.
- Malignancies other than SCLC within 5 years prior to randomization, with the exception of those with a negligible risk of metastasis or death (e.g., expected 5-year OS >90%) treated with expected curative outcome (such as adequately treated carcinoma in situ of the cervix, basal or squamous-cell skin cancer, localized prostate cancer treated surgically with curative intent, ductal carcinoma in situ treated surgically with curative intent)

- Women who are pregnant, lactating, or intending to become pregnant during the study
- History of autoimmune disease, including but not limited to myasthenia gravis, myositis, autoimmune hepatitis, systemic lupus erythematosus, rheumatoid arthritis, inflammatory bowel disease, vascular thrombosis associated with antiphospholipid syndrome, Wegener's granulomatosis, Sjögren's syndrome, Guillain-Barré syndrome, multiple sclerosis, vasculitis,or glomerulonephritis
    - Patients with a history of autoimmune-related hypothyroidism on thyroid replacement hormone therapy are eligible.
    - Patients with controlled Type I diabetes mellitus on an insulin regimen are eligible.
    - Patients with eczema, psoriasis, lichen simplex chronicus, or vitiligo with dermatologic manifestations only (e.g., patients with psoriatic arthritis would be excluded) are eligible for the study provided that they meet the following conditions:
        - Rash must cover less than 10% of body surface area
        - Disease is well controlled at baseline and only requires low potency topical steroids
        - No acute exacerbations of underlying condition within the last 12 months (not requiring psoralen plus ultraviolet A radiation [PUVA], methotrexate, retinoids, biologic agents, oral calcineurin inhibitors, high potency, or oral steroids)
- History of idiopathic pulmonary fibrosis, organizing pneumonia (e.g., bronchiolitis obliterans), drug-induced pneumonitis, idiopathic pneumonitis, or evidence of active pneumonitis on screening chest CT scan
    - History of radiation pneumonitis in the radiation field (fibrosis) is permitted.
- Positive test result for HIV
    - All patients must be tested for HIV; patients who test positive for HIV will be excluded.
- Patients with active hepatitis B (chronic or acute; defined as having a positive hepatitis B surface antigen [HBsAg] test result at screening) or hepatitis C virus (HCV)
    - Patients with past hepatitis B virus (HBV) infection or resolved HBV infection (defined as the presence of hepatitis B core antibody [HBcAb] and absence of HBsAg) are eligible. HBV DNA should be obtained in these patients prior to randomization.

    - Patients positive for HCV antibody are eligible only if PCR is negative for HCV RNA.
- Active tuberculosis
- Severe infections at the time of randomization, including but not limited to hospitalization for complications of infection, bacteremia, or severe pneumonia
- Significant cardiovascular disease, such as New York Heart Association cardiac disease (Class II or greater), myocardial infarction, or cerebrovascular accident within 3 months prior to randomization, unstable arrhythmias, or unstable angina
    - Patients with known coronary artery disease, congestive heart failure not meeting the above criteria, or left ventricular ejection fraction <50% must be on a stable medical regimen that is optimized in the opinion of the treating physician, in consultation with a cardiologist if appropriate.
- Major surgical procedure other than for diagnosis within 28 days prior to randomization or anticipation of need for a major surgical procedure during the course of the study
- Prior allogeneic bone marrow transplantation or solid organ transplant
- Any other diseases, metabolic dysfunction, physical examination finding, or clinical laboratory finding giving reasonable suspicion of a disease or condition that contraindicates the use of an investigational drug or that may affect the interpretation of the results or render the patient at high risk for treatment complications
- Patients with illnesses or conditions that interfere with their capacity to understand, follow, and/or comply with study procedures
- Treatment with any other investigational agent with therapeutic intent within 28 days prior to randomization
- Administration of a live, attenuated vaccine within 4 weeks before randomization or anticipation that such a live attenuated vaccine will be required during the study
    - Patients must not receive live, attenuated influenza vaccines (e.g., FluMist®) within 4 weeks prior to randomization, during treatment, and for 5 months following the last dose of atezolizumab/placebo.
- Prior treatment with CD137 agonists or immune checkpoint blockade therapies, anti–PD-1, and anti–PD-L1 therapeutic antibodies
- Treatment with systemic immunosuppressive medications (including, but not limited to corticosteroids, cyclophosphamide, azathioprine, methotrexate, thalidomide, and anti-tumor necrosis factor [anti-TNF] agents) within 1 week prior to randomization
    - Patients who have received acute systemic immunosuppressant medications (e.g., use of corticosteroids for nausea, vomiting, or management of or

premedication for allergic reactions) may be enrolled in the study after discussion with and approval by the Medical Monitor. In those patients, the need and length of the washout period prior to randomization will also be established in conjunction with the Medical Monitor.

  - The use of inhaled corticosteroids for chronic obstructive pulmonary disease, mineralocorticoids (e.g., fludrocortisone) for patients with orthostatic hypotension, and low-dose supplemental corticosteroids for adrenocortical insufficiency are allowed.

- History of severe allergic, anaphylactic, or other hypersensitivity reactions to chimeric or humanized antibodies or fusion proteins
- Known hypersensitivity or allergy to biopharmaceuticals produced in Chinese hamster ovary cells or any component of the atezolizumab formulation
- History of allergic reactions to carboplatin or etoposide

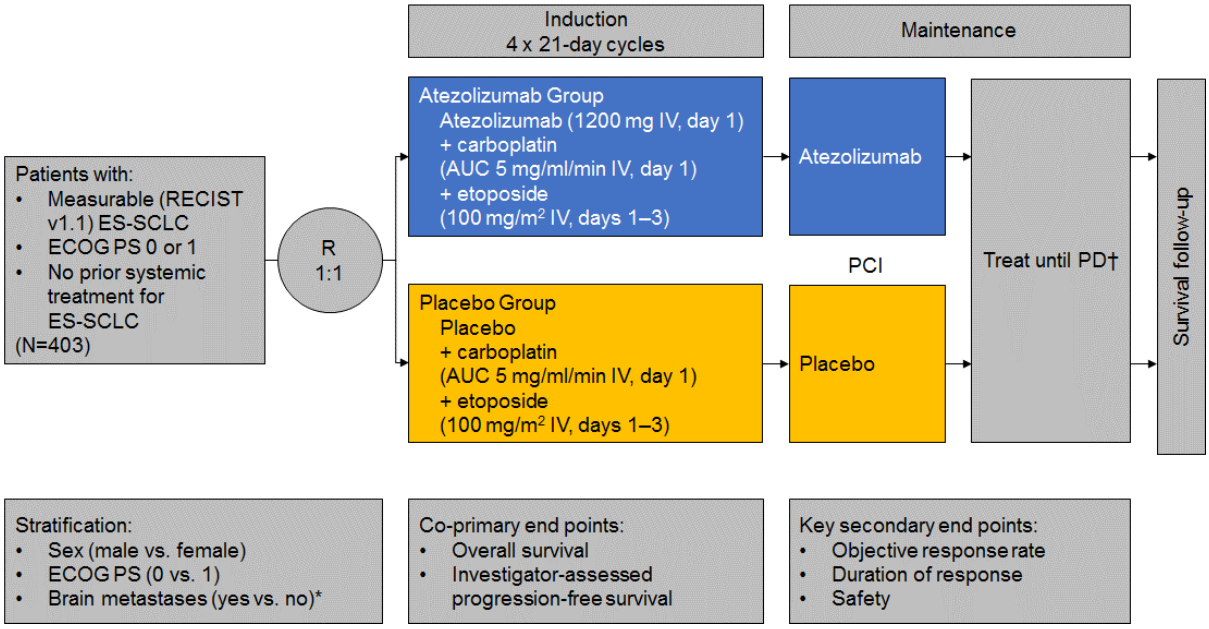
Predefined criteria for continuation of therapy beyond evidence of disease progression

Treatment should be discontinued in all patients (in both treatment groups) who exhibit evidence of disease progression per RECIST v1.1. However, the better to accommodate standard clinical practice (which is guided by the fact that patients with extensive-stage small-cell lung cancer [ES-SCLC] whose disease progresses after first-line treatment have limited treatment options and such options have limited efficacy and significant toxicity), patients may be considered for treatment beyond radiographic disease progression per RECIST, at the discretion of the investigator and after appropriate discussion with the patient and obtaining informed consent, only if all of the following criteria are met:

- Evidence of clinical benefit as assessed by the investigator
- No decline in Eastern Cooperative Oncology Group performance status that can be attributed to disease progression
- Absence of tumor progression at critical anatomical sites (e.g., leptomeningeal disease) that cannot be managed by protocol-allowed medical interventions
- Patients must provide written consent to acknowledge deferring other treatment options in favor of continuing study treatment at the time of initial progression

Patients who continue treatment beyond radiographic disease progression per RECIST should be closely monitored clinically and with a follow-up scan in 6 weeks or sooner if symptomatic deterioration occurs. Treatment should be discontinued if clinical deterioration due to disease progression occurs at any time, or if persistent disease growth is confirmed in a follow-up scan. In addition, patients should be discontinued for unacceptable toxicity or for any other signs or symptoms of deterioration attributable to disease progression as determined by the investigator after an integrated assessment of radiographic data and clinical status.

Figure S1. Trial Design.

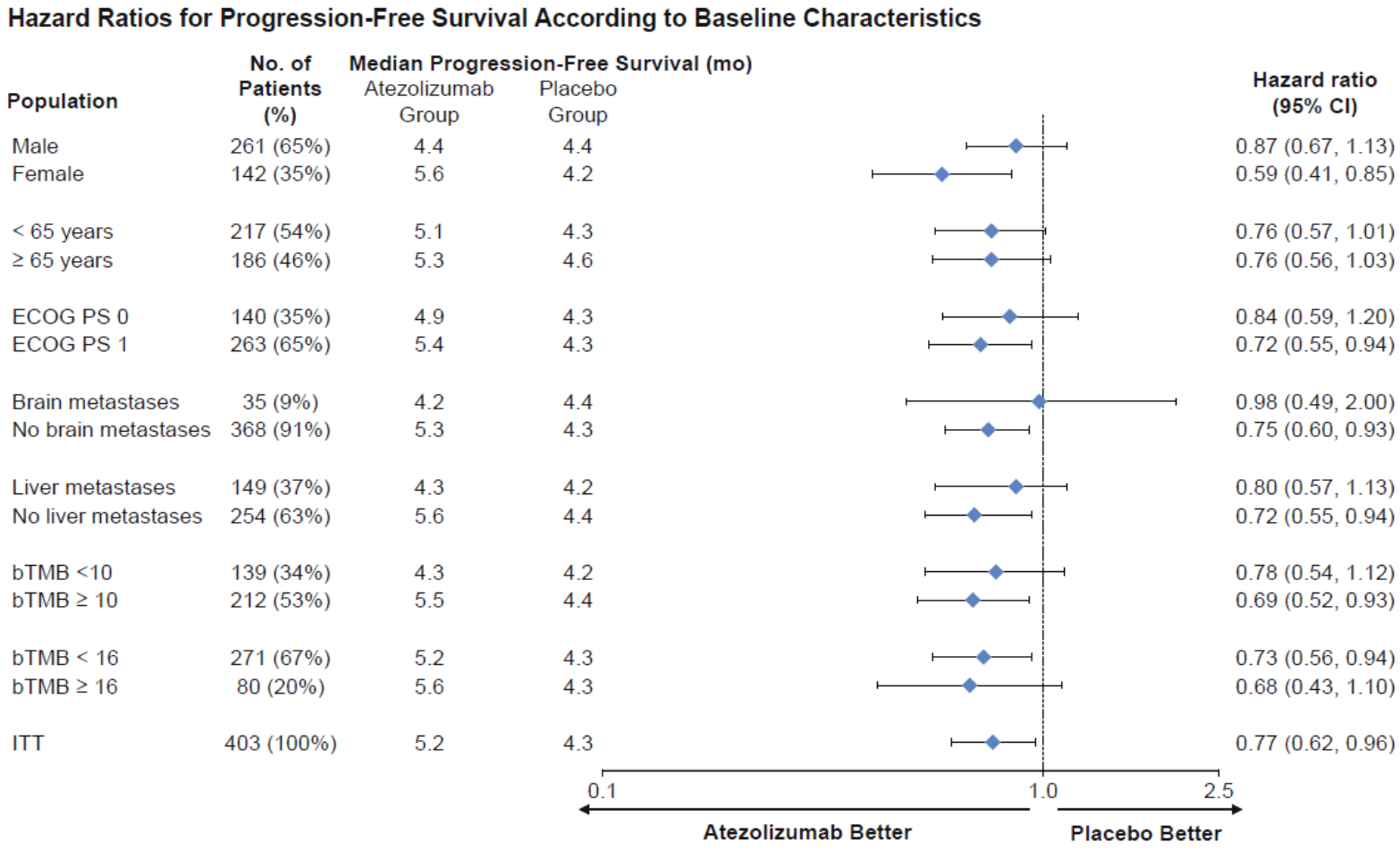


* Only patients with treated brain metastases were eligible; due to the low number of patients with brain metastases, the presence of brain metastases was excluded as a stratification factor in stratified data analyses.

† Patients who met prespecified criteria (see Supplementary Methods) were allowed to be treated beyond disease progression per RECIST v1.1 criteria until loss of clinical benefit in a blinded fashion.

PCI, prophylactic cranial irradiation; PD, disease progression.

Figure S2. Subgroup analysis of investigator-assessed progression-free survival according to baseline characteristics.



bTMB, blood tumor mutational burden; ECOG PS, Eastern Cooperative Oncology Group performance status.

Table S1. Veterans Administration Lung Study Group (VALG) staging system.[1]

Stage	Characteristics
Limited small-cell lung cancer	• Disease confined to one hemithorax, although local extensions may be present; • No extrathoracic metastases except for possible ipsilateral, supraclavicular nodes if they can be included in the same portal as the primary tumor; and • Primary tumor and regional nodes that can be adequately treated and totally encompassed in every portal
Extensive small-cell lung cancer	• Inoperable patients who cannot be classified as having limited disease

Table S2. Baseline Characteristics of All Enrolled Patients (Intention-to-Treat Population).*

Characteristic	Atezolizumab Group (N=201)	Placebo Group (N=202)
Median age (range) — yr	64 (28–90)	64 (26–87)
Age group — no. (%)		
<65 yr	111 (55.2)	106 (52.5)
65–74 yr	71 (35.3)	74 (36.6)
75–84 yr	18 (9.0)	21 (10.4)
≥85 yr	1 (0.5)	1 (0.5)
Male sex (eCRF) — no. (%)	129 (64.2)	132 (65.3)
Race or ethnic group — no. (%)		
White	163 (81.1)	159 (78.7)
Asian	33 (16.4)	36 (17.8)
Black	1 (0.5)	2 (1.0)
American Indian or Alaska Native	0	1 (0.5)
Unknown	4 (2.0)	4 (2.0)
Region		
Asia-Pacific	40 (19.9)	40 (19.8)
Europe and Middle East	116 (57.7)	107 (53.0)
North America	39 (19.4)	51 (25.2)
South America	6 (3.0)	4 (2.0)
ECOG performance status score (eCRF) — no. (%)		
0	73 (36.3)	67 (33.2)
1	128 (63.7)	135 (66.8)
History of tobacco use — no. (%)		

Characteristic	Atezolizumab Group (N=201)	Placebo Group (N=202)
Never	9 (4.5)	3 (1.5)
Current	74 (36.8)	75 (37.1)
Previous	118 (58.7)	124 (61.4)
Brain metastasis at enrollment (eCRF) — no. (%)	17 (8.5)	18 (8.9)
Liver metastases at enrollment — no. (%)	77 (38.3)	72 (35.6)
bTMB biomarker expression		
<10	71 (41.0)	68 (38.2)
≥10	102 (59.0)	110 (61.8)
<16	133 (76.9)	138 (77.5)
≥16	40 (23.1)	40 (22.5)
Median SLD (range) at baseline	113.0 (12.0–325.0)	105.5 (15.0–353.0)
Previous anticancer treatments		
Chemotherapy/non-anthracycline†	8 (4.0)	12 (5.9)
Prior radiotherapy	25 (12.4)	28 (13.9)
Prior cancer-related surgery	33 (16.4)	25 (12.4)

* The date of data cutoff was April 24, 2018.

† Cisplatin, etoposide, plus concurrent radiation (atezolizumab group, n=6; placebo group, n=7); carboplatin, etoposide, plus concurrent radiation (atezolizumab group, n=2; placebo group, n=6).

bTMB denotes blood tumor mutational burden per megabase pair; ECOG, Eastern Cooperative Oncology Group; eCRF, electronic case report form; SLD, sum of longest diameter of target lesions.

Table S3. Sites of metastases.

	Atezolizumab Group (N=201)	Placebo Group (N=202)
Patients with metastatic site — no. (%)		
Lung	177 (88.1)	172 (85.1)
Lymph nodes	160 (79.6)	168 (83.2)
Liver	77 (38.3)	72 (35.6)
Pleura	52 (25.9)	53 (26.2)
Bone	39 (19.4)	42 (20.8)
Adrenal gland	37 (18.4)	35 (17.3)
Brain	17 (8.5)	18 (8.9)
Mediastinum	15 (7.5)	16 (7.9)
Pancreas	8 (4.0)	9 (4.5)
Chest	6 (3.0)	10 (5.0)
Pericardial cavity	9 (4.5)	5 (2.5)
Kidney	6 (3.0)	4 (2.0)
Peritoneum	4 (2.0)	3 (1.5)
Soft tissue	4 (2.0)	2 (1.0)
Skin	1 (0.5)	2 (1.0)
Trachea	3 (1.5)	0
Neck	0	2 (1.0)
Breast	0	1 (0.5)
Bronchus	0	1 (0.5)
Esophagus	0	1 (0.5)
Ovary	1 (0.5)	0
Pelvis	1 (0.5)	0
Other	20 (10.0)	11 (5.4)

Table S4. Subsequent Cancer Therapies.

	Atezolizumab Group (N=201)	Placebo Group (N=202)
Line of therapy		
Second	101 (50.2)	116 (57.4)
Third	29 (14.4)	38 (18.8)
Fourth	3 (1.5)	15 (7.4)
Therapy type		
Total number of patients with at least one treatment	104 (51.7)	116 (57.4)
Total number of treatments — no.	138	176
Chemotherapy/non-anthracycline	81 (40.3)	88 (43.6)
Chemotherapy/anthracycline	31 (15.4)	46 (22.8)
Immunotherapy	6 (3.0)	15 (7.4)
Other	2 (1.0)	2 (1.0)
Targeted therapy	2 (1.0)	1 (0.5)

Data are no. of patients with at least one treatment (%) unless otherwise specified.

Multiple cases within a specific line of therapy and regimen for a patient were counted once for the frequency of line of therapy or regimen name. A patient was counted more than once if that patient received more than one therapy type under each line and regimen.

The data cutoff date was April 24, 2018.

Table S5. Confirmed Investigator-Assessed Objective Response Rate and Duration of Response (Intention-to-Treat Population with Measurable Disease at Baseline).*

Variable	Atezolizumab Group	Placebo Group
Response†		
No. of evaluable patients	201	202
Objective confirmed response — no. (%)	121 (60.2 [53.1–67.0])	130 (64.4 [57.3–71.0])
Complete	5 (2.5 [0.8–5.7])	2 (1.0 [0.1–3.5])
Partial	116 (57.7 [50.6–64.6])	128 (63.4 [56.3–70.0])
Stable disease — no. (% [95% CI])	42 (20.9 [15.5–27.2])	43 (21.3 [15.9–27.6])
Progressive disease — no. (% [95% CI])	22 (10.9 [7.0–16.1])	14 (6.9 [3.8–11.4])
Patients with missing data or who could not be evaluated — no. (%)	16 (8.0)	15 (7.4)
Duration of response‡		
No. of evaluable patients	121	130
Median duration of response (range) — mo	4.2 (1.4+–19.5)	3.9 (2.0–16.1+)
Patients with ongoing response at data cutoff date — no. (%)	18 (14.9)	7 (5.4)

* The date of data cutoff was April 24, 2018.

† Objective response was defined as complete response or partial response as determined by the investigator according to RECIST v1.1 criteria.

‡ Duration of response was assessed among patients who had an objective response and was defined as the time from first occurrence of a documented objective response to the time of disease progression as determined by the investigator using RECIST or death from any cause, whichever occurred first.

+ denotes a censored observation; CI denotes confidence interval.

Table S6. Treatment Exposure.

	Atezolizumab Group (N=198)			Placebo Group (N=196)		
	Atezolizumab	**Carboplatin**	**Etoposide**	**Placebo**	**Carboplatin**	**Etoposide**
Median treatment duration						
Median — months	4.7	2.3	2.3	4.1	2.2	2.2
0–3 months — no. (%)	47 (23.7)	193 (97.5)	191 (96.5)	41 (20.9)	191 (97.4)	191 (97.4)
3–6 months — no. (%)	87 (43.9)	5 (2.5)	7 (3.5)	113 (57.7)	5 (2.6)	5 (2.6)
6–12 months — no. (%)	41 (20.7)	0	0	30 (15.3)	0	0
>12 months — no. (%)	23 (11.6)	0	0	12 (6.1)	0	0
Median dose intensity — %	94.9	92.3	89.4	94.7	93.3	90.3
Median doses — no.	7	4	12	6	4	12
Total cumulative dose — mg						
Mean (SD)	10193.0 (7166.6)	2019.2 (642.2)	1965.8 (539.8)	0	2145.7 (645.0)	2034.5 (477.2)
Median	8400.0	2062.5	2055.2	0	2175	2131.7

Dose intensity is the number of doses actually received divided by the expected number of doses.

The data cutoff date was April 24, 2018.

Table S7. Safety Summary.

Patients — no. (%)	**Atezolizumab Group (N=198)**	**Placebo Group (N=196)**
Patients with ≥1 AE	198 (100)	189 (96.4)
Grade 3–4 AEs	133 (67.2)	125 (63.8)
Grade 5 AEs	4 (2.0)	11 (5.6)
Treatment-related AEs*	188 (94.9)	181 (92.3)
Treatment-related Grade 3–4 AEs	112 (56.6)	110 (56.1)
Treatment-related Grade 5 AEs	3 (1.5)	3 (1.5)
Serious AEs	74 (37.4)	68 (34.7)
Treatment-related serious AEs*	45 (22.7)	37 (18.9)
AEs leading to withdrawal from any treatment*	22 (11.1)	6 (3.1)
AEs leading to withdrawal from carboplatin	5 (2.5)	1 (0.5)
AEs leading to withdrawal from etoposide	8 (4.0)	2 (1.0)

* Incidence of treatment-related AEs, serious treatment-related AEs, and AEs leading to withdrawal from any treatment are for any treatment component. The date of data cutoff was April 24, 2018. Multiple occurrences of the same AE in one patient were counted once at the highest grade for the preferred term.

AE denotes adverse event.

Table S8. Treatment-Related Adverse Events.*

	Atezolizumab Group (N=198)			Placebo Group (N=196)		
Patients — no. (%)	**Grade 1–2**	**Grade 3–4**	**Grade 5**	**Grade 1–2**	**Grade 3–4**	**Grade 5**
Treatment-related AEs	73 (36.9)	112 (56.6)	3 (1.5)	68 (34.7)	110 (56.1)	3 (1.5)
Treatment-related AEs with an incidence of ≥10% in any arm, grade 3–4 severity with incidence of ≥1% in any arm, or grade 5 severity						
Neutropenia	26 (13.1)	45 (22.7)	1 (0.5)	20 (10.2)	48 (24.5)	0
Anemia	49 (24.7)	28 (14.1)	0	41 (20.9)	24 (12.2)	0
Alopecia	69 (34.8)	0	0	66 (33.7)	0	0
Nausea	62 (31.3)	1 (0.5)	0	58 (29.6)	1 (0.5)	0
Fatigue	39 (19.7)	3 (1.5)	0	37 (18.9)	1 (0.5)	0
Neutrophil count decreased	7 (3.5)	28 (14.1)	0	12 (6.1)	33 (16.8)	0
Decreased appetite	39 (19.7)	2 (1.0)	0	26 (13.3)	0	0
Thrombocytopenia	12 (6.1)	20 (10.1)	0	14 (7.1)	15 (7.7)	0
Platelet count decreased	17 (8.6)	7 (3.5)	0	21 (10.7)	7 (3.6)	0
Vomiting	25 (12.6)	2 (1.0)	0	19 (9.7)	3 (1.5)	0
Constipation	19 (9.6)	1 (0.5)	0	25 (12.8)	0	0
Leukopenia	15 (7.6)	10 (5.1)	0	10 (5.1)	8 (4.1)	0
White blood cell count decreased	10 (5.1)	6 (3.0)	0	16 (8.2)	9 (4.6)	0
Diarrhea	15 (7.6)	4 (2.0)	0	18 (9.2)	1 (0.5)	0
Asthenia	14 (7.1)	3 (1.5)	0	12 (6.1)	2 (1.0)	0
Febrile neutropenia	0	6 (3.0)	0	0	12 (6.1)	0
Infusion-related reaction	6 (3.0)	4 (2.0)	0	9 (4.6)	1 (0.5)	0
Hypomagnesemia	7 (3.5)	0	0	5 (2.6)	2 (1.0)	0

	Atezolizumab Group (N=198)			Placebo Group (N=196)		
Patients — no. (%)	**Grade 1–2**	**Grade 3–4**	**Grade 5**	**Grade 1–2**	**Grade 3–4**	**Grade 5**
Peripheral neuropathy	4 (2.0)	2 (1.0)	0	4 (2.0)	0	0
Hypokalemia	2 (1.0)	0	0	4 (2.0)	2 (1.0)	0
Pneumonia	1 (0.5)	3 (1.5)	1 (0.5)	2 (1.0)	0	1 (0.5)
Pneumonitis	2 (1.0)	1 (0.5)	0	2 (1.0)	2 (1.0)	0
Pancytopenia	1 (0.5)	1 (0.5)	0	1 (0.5)	3 (1.5)	0
Acute kidney injury	2 (1.0)	2 (1.0)	0	1 (0.5)	0	0
Lung infection	1 (0.5)	0	0	0	2 (1.0)	0
Cardiopulmonary failure	0	0	0	0	0	1 (0.5)
Death	0	0	1 (0.5)	0	0	0
Septic shock	0	0	0	0	0	1 (0.5)

* Incidence of treatment-related adverse events for any treatment. The date of data cutoff was April 24, 2018. Multiple occurrences of the same AE in one patient were counted once at the highest grade for the preferred term.

AE denotes adverse event.

Table S9. Serious Treatment-Related Adverse Events.*

	Atezolizumab Group (N=198)			Placebo Group (N=196)		
Patients — no. (%)	**Grade 1–2**	**Grade 3–4**	**Grade 5**	**Grade 1–2**	**Grade 3–4**	**Grade 5**
Neutropenia	0	6 (3.0)	1 (0.5)	0	8 (4.1)	0
Febrile neutropenia	0	4 (2.0)	0	0	9 (4.6)	0
Thrombocytopenia	0	5 (2.5)	0	0	4 (2.0)	0
Pneumonia	0	3 (1.5)	1 (0.5)	0	0	1 (0.5)
Anemia	0	3 (1.5)	0	0	2 (1.0)	0
Pancytopenia	0	0	0	1 (0.5)	3 (1.5)	0
Vomiting	0	2 (1.0)	0	0	2 (1.0)	0
Diarrhea	1 (0.5)	2 (1.0)	0	0	0	0
Leukopenia	0	2 (1.0)	0	0	1 (0.5)	0
Infusion-related reaction	0	1 (0.5)	0	2 (1.0)	0	0
Pneumonitis	0	1 (0.5)	0	0	2 (1.0)	0
Lung infection	0	0	0	0	2 (1.0)	0
Platelet count decreased	0	0	0	1 (0.5)	1 (0.5)	0
Acute kidney injury	0	2 (1.0)	0	0	0	0
Asthenia	0	2 (1.0)	0	0	0	0
Autoimmune thyroiditis	2 (1.0)	0	0	0	0	0
Death	0	0	1 (0.5)	0	0	0
Cardiopulmonary failure	0	0	0	0	0	1 (0.5)
Septic shock	0	0	0	0	0	1 (0.5)
Acute pancreatitis	0	1 (0.5)	0	0	0	0

	Atezolizumab Group (N=198)			Placebo Group (N=196)		
Patients — no. (%)	**Grade 1–2**	**Grade 3–4**	**Grade 5**	**Grade 1–2**	**Grade 3–4**	**Grade 5**
Atrioventricular block complete	0	1 (0.5)	0	0	0	0
Colitis	0	1 (0.5)	0	0	0	0
Dehydration	0	1 (0.5)	0	0	0	0
Fatigue	0	1 (0.5)	0	0	0	0
Ileus	0	1 (0.5)	0	0	0	0
Jaundice	0	1 (0.5)	0	0	0	0
Liver function test increased	0	1 (0.5)	0	0	0	0
Lower respiratory tract infection	0	1 (0.5)	0	0	0	0
Nausea	0	1 (0.5)	0	0	0	0
Peripheral neuropathy	0	1 (0.5)	0	0	0	0
Pulmonary edema	0	1 (0.5)	0	0	0	0
Skin toxicity	0	1 (0.5)	0	0	0	0
Transaminases increased	0	1 (0.5)	0	0	0	0
Trigeminal neuralgia	0	1 (0.5)	0	0	0	0
Tubulointerstitial nephritis	0	1 (0.5)	0	0	0	0
Hypokalemia	0	0	0	0	1 (0.5)	0
Hypomagnesemia	0	0	0	0	1 (0.5)	0
Neutropenic sepsis	0	0	0	0	1 (0.5)	0
Neutrophil count decreased	0	0	0	0	1 (0.5)	0
Pancreatitis	0	0	0	0	1 (0.5)	0
Urinary tract infection	0	0	0	0	1 (0.5)	0

	Atezolizumab Group (N=198)			Placebo Group (N=196)		
Patients — no. (%)	**Grade 1–2**	**Grade 3–4**	**Grade 5**	**Grade 1–2**	**Grade 3–4**	**Grade 5**
White blood cell count decreased	0	0	0	0	1 (0.5)	0
Autoimmune colitis	1 (0.5)	0	0	0	0	0
Blood creatinine increased	1 (0.5)	0	0	0	0	0
Bronchitis	1 (0.5)	0	0	0	0	0
Cytomegalovirus infection	1 (0.5)	0	0	0	0	0
Diverticular perforation	1 (0.5)	0	0	0	0	0
Guillain−Barre syndrome	0	1 (0.5)	0	0	0	0
Hemoptysis	1 (0.5)	0	0	0	0	0
Pleural effusion	1 (0.5)	0	0	0	0	0

* Incidence of treatment-related adverse events for any treatment. The date of data cutoff was April 24, 2018. Multiple occurrences of the same AE in one patient were counted once at the highest grade for the preferred term.

AE denotes adverse event.

Table S10. Immune-Related Adverse Events.

Patients — no. (%)	**Atezolizumab Group (N=198)**	**Placebo Group (N=196)**
Rash		
All grades	37 (18.7)	20 (10.2)
Grade 3–4	4 (2.0)	0
Hypothyroidism		
All grades	25 (12.6)	1 (0.5)
Grade 3–4	0	0
Hepatitis (diagnosis)		
All grades	14 (7.1)	9 (4.6)
Grade 3–4	3 (1.5)	0
Hepatitis (laboratory abnormalities)		
All grades	14 (7.1)	9 (4.6)
Grade 3–4	3 (1.5)	0
Infusion-related reaction		
All grades	11 (5.6)	10 (5.1)
Grade 3–4	4 (2.0)	1 (0.5)
Hyperthyroidism		
All grades	11 (5.6)	5 (2.6)
Grade 3–4	0	0
Pneumonitis		
All grades	4 (2.0)	5 (2.6)
Grade 3–4	1 (0.5)	2 (1.0)
Colitis		
All grades	3 (1.5)	0
Grade 3–4	2 (1.0)	0
Pancreatitis		
All grades	1 (0.5)	2 (1.0)
Grade 3–4	1 (0.5)	2 (1.0)
Severe cutaneous reaction		
All grades	2 (1.0)	0
Grade 3–4	0	0
Adrenal insufficiency		
All grades	0	2 (1.0)
Grade 3–4	0	0
Rhabdomyolysis		
All grades	2 (1.0)	0
Grade 3–4	1 (0.5)	0
Nephritis		
All grades	1 (0.5)	1 (0.5)
Grade 3–4	1 (0.5)	0
Hypophysitis		
All grades	1 (0.5)	0
Grade 3–4	0	0
Vasculitis		
All grades	0	1 (0.5)
Grade 3–4	0	0

Patients — no. (%)	**Atezolizumab Group (N=198)**	**Placebo Group (N=196)**
Diabetes mellitus		
All grades	1 (0.5)	0
Grade 3–4	0	0
Guillain−Barre Syndrome		
All grades	1 (0.5)	0
Grade 3–4	1 (0.5)	0

Immune-related AEs were defined using MedDRA Preferred Terms that included both diagnosed immune conditions and signs and symptoms potentially representative of immune-related events, regardless of investigator-assessed causality.

The data cutoff date was April 24, 2018.

AE denotes adverse event.

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